

Forklift-Warriors

Team

Landscape

Opportunities

Timeline

The Team

Hank Davidson

Lead, VR Dev, Music/Sound Development

Jerry Denning

VR Development

Steven Drain

VR Development

Kenzie Burt

Lead 3D Artist

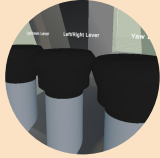
Jaden Rogers

3D Artist

Our forklift simulation is a fun and engaging way to make learning forklifts easy and accessible with the help of VR

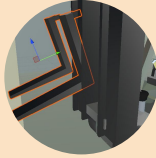
Stevens' Contributions

VR Developer



Forklift Levers

Lever functionality for up/down, left/right, and angled



Spoke Movement

Moves up, down left right, and angled



NPC

Moving NPC from A to B like a real guy does



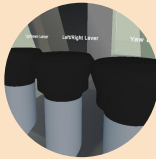
Ragdoll

Added ragdolls to NPC for vehicle collision



Objective

Spawns boxes and objective points around the map



Subtitle/VoiceOver (Lever)

When specific settings enabled, voiceover and subtitles play

Code Snippets

- Calls each lever and applies movement to the spoke object
- When you stop grabbing the lever it stops the spoke where it is

```
3 references
float GetLeverInput(XRLever lever)
{
    if (!lever.isSelected) return 0f;
    return ApplyDeadzone(lever.value);
}

1 reference
void RaisingFork()
{
    float input = GetLeverInput(raiseLever);
    if (input != 0f)
    {
        Vector3 pos = forkTransform.localPosition;
        pos.y += input * forkRaiseSpeed * Time.deltaTime;
        pos.y = Mathf.Clamp(pos.y, restPosition.y + minY, restPosition.y + maxY);
        forkTransform.localPosition = pos;
    }
}

1 reference
void LeftRightFork()
{
    float input = GetLeverInput(leftRightLever);
    if (input != 0f)
    {
        Vector3 pos = forkTransform.localPosition;
        pos.x += input * forkSlideSpeed * Time.deltaTime;
        pos.x = Mathf.Clamp(pos.x, restPosition.x + minX, restPosition.x + maxX);
        forkTransform.localPosition = pos;
    }
}

1 reference
void TiltFork()
{
    float input = GetLeverInput(tiltLever);
    if (input != 0f)
    {
        currentTilt += input * forkTiltSpeed * Time.deltaTime;
        currentTilt = Mathf.Clamp(currentTilt, minTilt, maxTilt);
        forkTransform.localRotation = restRotation * Quaternion.Euler(currentTilt, 0f, 0f);
    }
}
```

Code Snippets

- Grabs the box and snaps it to its end spot
- updates the score to spawn the next palette and objective

```
1 reference
void OnTriggerEnter(Collider other)
{
    if (other.transform.root != other.transform && other.CompareTag("boxpallet"))
        return;

    if (other.CompareTag("boxpallet") && !isSnapped)
    {
        Rigidbody rb = other.GetComponent<Rigidbody>();
        if (rb != null)
        {
            objectiveAreaVisuals.SetActive(false);
            //play animation like confetti or something
            Snap(other.transform, rb);
        }
    }
}

1 reference
private void Snap(Transform root, Rigidbody rb)
{
    boxPallet = rb;
    boxPallet.isKinematic = true;
    root.SetParent(snapPoint.transform);
    root.position = snapPoint.transform.position;
    root.rotation = snapPoint.transform.rotation;
    isSnapped = true;

    foreach (Collider col in root.GetComponentsInChildren<Collider>())
    {
        col.enabled = false;
    }

    Debug.Log("Snapped permanently: " + root.name);
    scoreScript.ScoreUpdated();
}
```

Hank

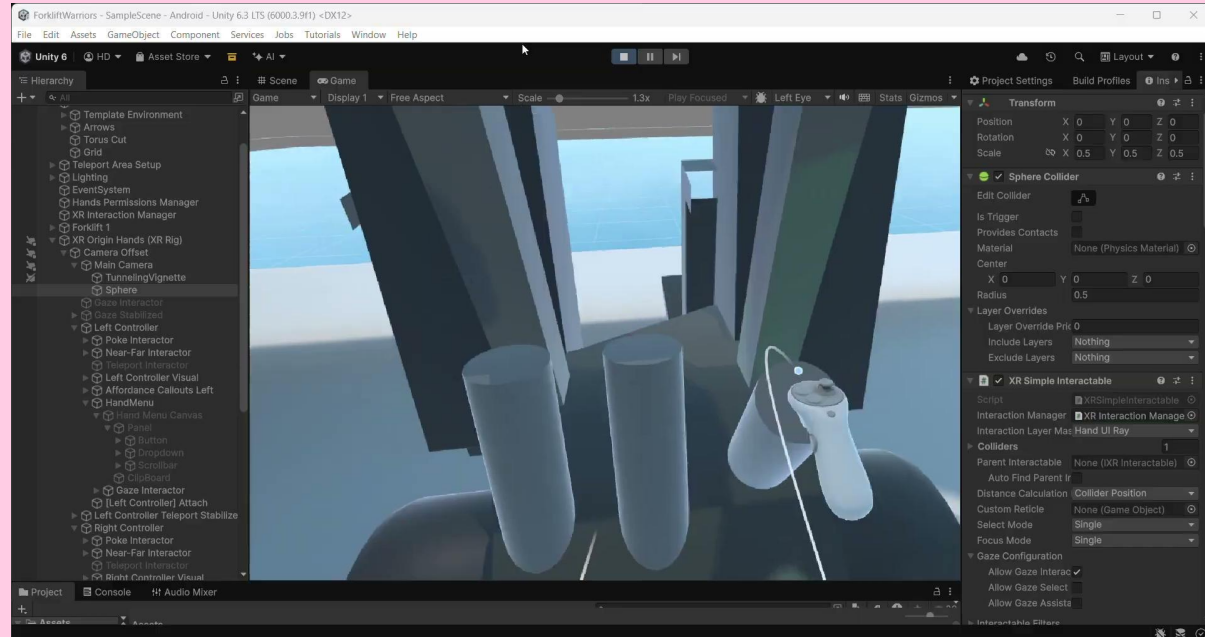
Created the hand menu and audio for our UX experience!

- Hand Menu included:

- An interactive VR-friendly in-game navigator
- Accessibility feature manager

- Audio

- Created the environmental audio effects to deepen the immersion
- Recorded voice que's when the worker is hit



Audio



Forklift Idle

Sourced from real forklift sounds online, the system is set up to automatically switch from idle to driving mode in the audio manager



Warehouse Ambience

Used Epidemic sound and youtube to mix down an immersive warehouse soundscape



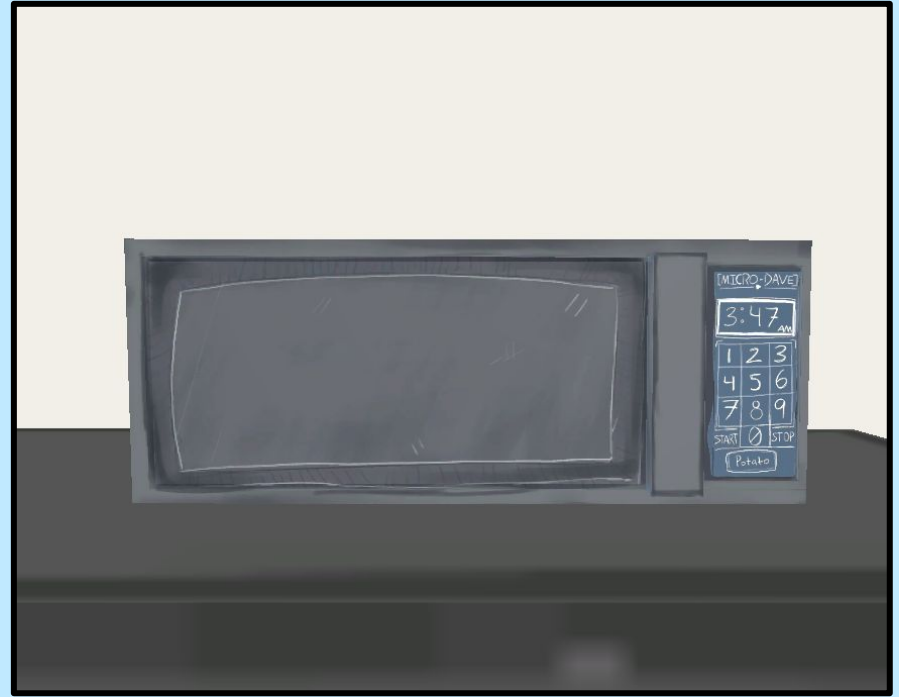
Music

Sampled music from Epidemic Sound, a free software to pull loops and samples to use in games and projects

KENZIE

MePaint

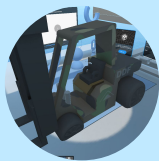
TEXTURES



Jaden's Contributions

Forklift Model

3D Model & Rig

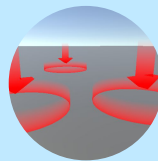


My biggest contribution to this project, I was responsible for the model, texturing, and rigging of the forklift, which is the main hero asset of the game.

Final implementation of the forklift was handled by the rest of the team, where the rig and handles were done procedurally instead of authoring pre-baked animations

Objective Markers

3D Model & Shader



One of the accessibility features of our game was the inclusion of optional objective markers designed to reduce cognitive load on the user.

The markers include two models, a ring and an arrow, with custom UVs for a nice fading effect from the custom shader I wrote for this object specifically.

High Contrast UI

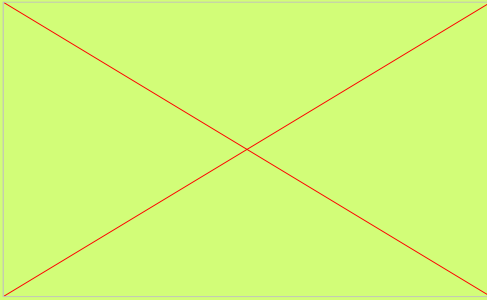
Accessibility System



I created a script that swaps out the clipboards UI for a high-contrast variant designed to help improve readability for users with vision disabilities.

This script runs dynamically, allowing the user to toggle this mode during gameplay if desired.

Jerry



Forklift Controls

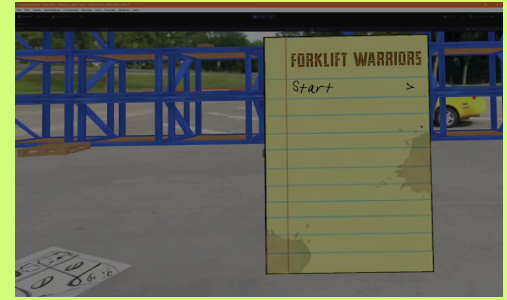
I programmed the main vehicle script for actually driving the forklift. I also used group member feedback to get make the driving more realistic.



Bug Fixes/Touch-up

I also ironed out some bugs. For example, I fixed the NavMesh surfaces so that NPCs don't walk on walls.

I also worked on touching up the game. I applied all of Kenzie's texturing to all of the models in the game. I also worked on a little bit of the graphics (a bit brighter).



Start Menu

I created the entire starting menu, and added all of the features to the starting menu using our 3D assets and fonts I found.

ACCESSIBILITY

Audio

- hover over the lever; it tells you what it is
- Designed for people who are hard of hearing, this feature allows an individual to better learn and understand their environment.

Cognitive

- Markers pointing to key objective areas
- Designed to allow users to understand waypoints in a warehouse that are of importance. In our case to guide users on where to place boxes

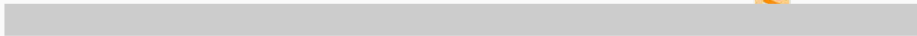
Visual

- High Contrast Menu system
- Helps people read the menus easier without much strain.

Motor

- Level long-distance usage (toggle)
- Designed for the user to have ease of access to our lever controls. Someone with limb immobility can easily access our game and its controls

DEMO TIME



Thank you!

JUST IN CASE

